

EFPB Research Pipeline

Friend-run installation and experiment instructions — release 0.1.0 scaffold

Important status: this package includes a deterministic surrogate backend and a real SUMO/TraCI emergency runner. The emergency runner uses SUMO for vehicle movement and signal execution, while pedestrians remain a reproducible virtual queue because the bundled legacy network has no validated pedestrian walking areas. Read SUMO_EMERGENCY_SCOPE.md and do not present this limited pilot as complete multimodal evidence.

1. Install

From the extracted package directory:

```
cd efpb-research-pipeline
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e '[dev,analysis]'
```

Verify the code:

```
python -m pytest -q
```

Expected result: 5 tests pass.

2. SUMO setup and network check

Install SUMO using the supported package for your operating system. Then set SUMO HOME and PATH:

```
export SUMO_HOME=/path/to/sumo
export PATH="$SUMO_HOME/bin:$PATH"
bash sumo/build_network.sh
PYTHONPATH=src python validate_sumo.py
```

If validation fails, stop. Check the SUMO version, environment variables, permissions, and the generated network. Manually inspect the network in sumo-gui before any confirmatory use.

Emergency real-SUMO run (one-day deadline)

```
PYTHONPATH=src python run_sumo_actual.py --config configs/experiments/sumo_emergency.yaml --scenario
S1_balanced --controller efpb --seed 9001
PYTHONPATH=src python run_sumo_batch.py --config configs/experiments/sumo_emergency.yaml --machine 1
--machines 2
```

Run the same batch command on the second computer with --machine 2. The emergency matrix has 36 real SUMO runs (3 scenarios × 4 controllers × 3 paired seeds). Results are written to data/sumo_runs/. After copying both computers' run folders together, aggregate them with:

```
python aggregate_sumo.py --input data/sumo_runs --output data/processed/sumo_metrics.csv
python analyze_sumo.py --input data/processed/sumo_metrics.csv --output
data/processed/sumo_analysis.json
```

This is the narrow, time-boxed pilot defined in SUMO_EMERGENCY_SCOPE.md.

3. Smoke pipeline

```
python run_experiment.py --config configs/experiments/smoke.yaml
python aggregate.py --input data/runs --output data/processed/run_metrics.csv
python analyze.py --input data/processed/run_metrics.csv --output data/processed/analysis.json
python generate_figures.py --input data/processed/run_metrics.csv --output data/figures
python build_stimuli.py --input data/runs --output data/stimuli
```

This runs 2 scenarios × 4 controllers × 1 paired seed = 8 surrogate runs.

4. Calibration and benchmark

```
python calibrate.py --config configs/experiments/calibration.yaml
python benchmark.py --config configs/experiments/benchmark.yaml
python estimate_runtime.py
```

Inspect calibration output before freezing parameters. Do not tune confirmatory seeds.

5. Experiment matrix

Group	Runs	Definition
Confirmatory core	1,080	9 demand cells × 4 controllers × 30 paired seeds
Robustness	360	9 targeted scenarios × 4 controllers × 10 seeds
Ablations	100	5 cells × 2 controllers × 10 seeds
Total	1,540	Traffic runs

Run the traffic groups with:

```
python run_experiment.py --config configs/experiments/confirmatory_core.yaml
python run_experiment.py --config configs/experiments/robustness.yaml
python run_experiment.py --config configs/experiments/ablations.yaml
```

6. Two-computer split

Computer 1

```
python run_all.py --machine 1 --machines 2 --config configs/experiments/confirmatory_core.yaml --config
configs/experiments/robustness.yaml
```

Computer 2

```
python run_all.py --machine 2 --machines 2 --config configs/experiments/confirmatory_core.yaml --config
configs/experiments/robustness.yaml
```

Use the same package, configuration, SUMO release, and seed lists on both computers. Copy completed data/runs subdirectories to one computer, then aggregate once. Do not delete failed runs or rerun selectively without recording the reason.

7. Outputs

Each run is stored under data/runs/{scenario}/{controller}/{seed}/{run_id}/ and includes manifest.json, resolved_config.json, states.csv, decisions.jsonl, explanations.jsonl, events.jsonl, run_metrics.json, and run_summary.json. Aggregates go to data/processed/, figures to data/figures/, and stimuli to data/stimuli/.

8. Runtime estimate

Measured surrogate benchmark: 3.144 seconds for 660 simulated seconds on the development Mac. The provisional 1,540-run matrix extrapolates to about 8.56 CPU hours, or 4.28 ideal hours on two equal computers. Allow approximately 6.4–8.6 wall-clock hours for overhead. Replace this with a measured headless SUMO benchmark before scheduling final confirmatory runs.

9. Research integrity

The paper must not contain fabricated results. Surrogate output is engineering validation only. Report SUMO traffic findings only after SUMO/TraCI validation, calibration lock, complete runs, and result validation. HCI claims require the approved participant study; stimulus generation alone is not HCI evidence.

Source: EFPB research plan and repository release documentation.